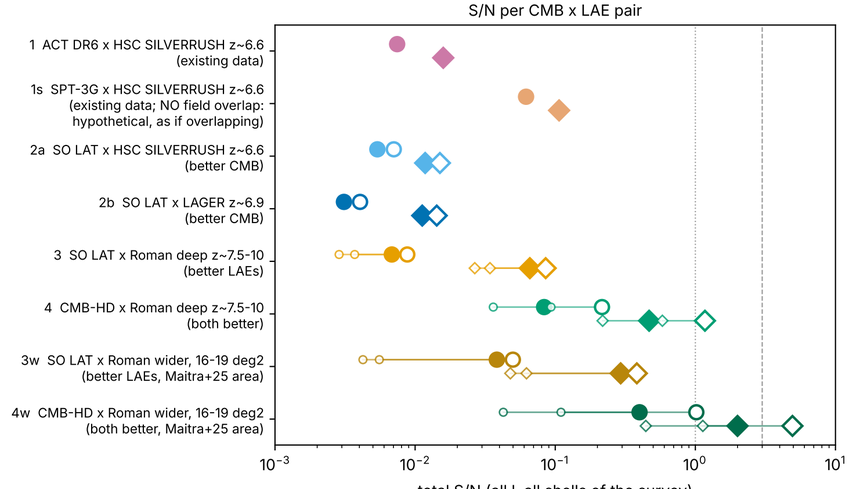
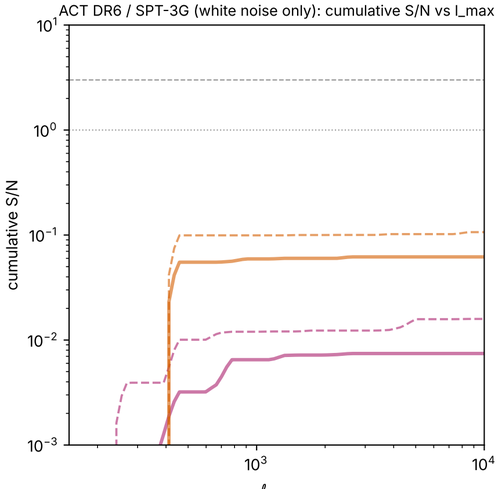
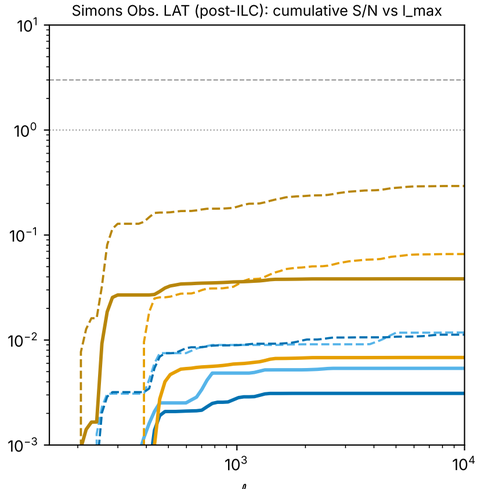
kSZ^2 x LAE detectability for the survey pairs: existing data, better CMB, better LAEs, both (fiducial reionisation history; real areas, counts and footprints)



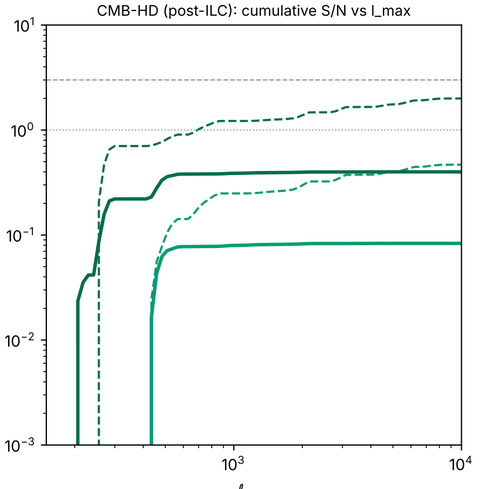
La Plante's galaxy model (validated; may be low by up to 2x)
simulated LAEs (excess not yet confirmed)
filled: post-ILC noise (ACT / SPT-3G: white noise only)
open: white (instrument) noise
small marker: Roman deep = z = 7.3 LF (observed decline); Roman wider = HLWAS deep-tier depth
1σ
3σ



1: ACT DR6 x HSC SILVERRUSH z~6.6, La Plante's galaxy model: 0.0074
1: ACT DR6 x HSC SILVERRUSH z~6.6, simulated LAEs: 0.016
1s: SPT-3G x HSC SILVERRUSH z~6.6 (no overlap; as if), La Plante's galaxy model: 0.062
1s: SPT-3G x HSC SILVERRUSH z~6.6 (no overlap; as if), simulated LAEs: 0.11
1σ
3σ



2a: SO LAT x HSC SILVERRUSH z~6.6, La Plante's galaxy model: 0.0054
2a: SO LAT x HSC SILVERRUSH z~6.6, simulated LAEs: 0.012
2b: SO LAT x LAGER z~6.9, La Plante's galaxy model: 0.0031
2b: SO LAT x LAGER z~6.9, simulated LAEs: 0.011
3: SO LAT x Roman deep z~7.5-10, La Plante's galaxy model: 0.0068
3: SO LAT x Roman deep z~7.5-10, simulated LAEs: 0.066
3w: SO LAT x Roman wider, 16-19 deg2, La Plante's galaxy model: 0.038
3w: SO LAT x Roman wider, 16-19 deg2, simulated LAEs: 0.29
1σ
3σ



4: CMB-HD x Roman deep z~7.5-10, La Plante's galaxy model: 0.083
4: CMB-HD x Roman deep z~7.5-10, simulated LAEs: 0.47
4w: CMB-HD x Roman wider, 16-19 deg2, La Plante's galaxy model: 0.4
4w: CMB-HD x Roman wider, 16-19 deg2, simulated LAEs: 2
1σ
3σ